# IP asoslari

## IP Fundamentals

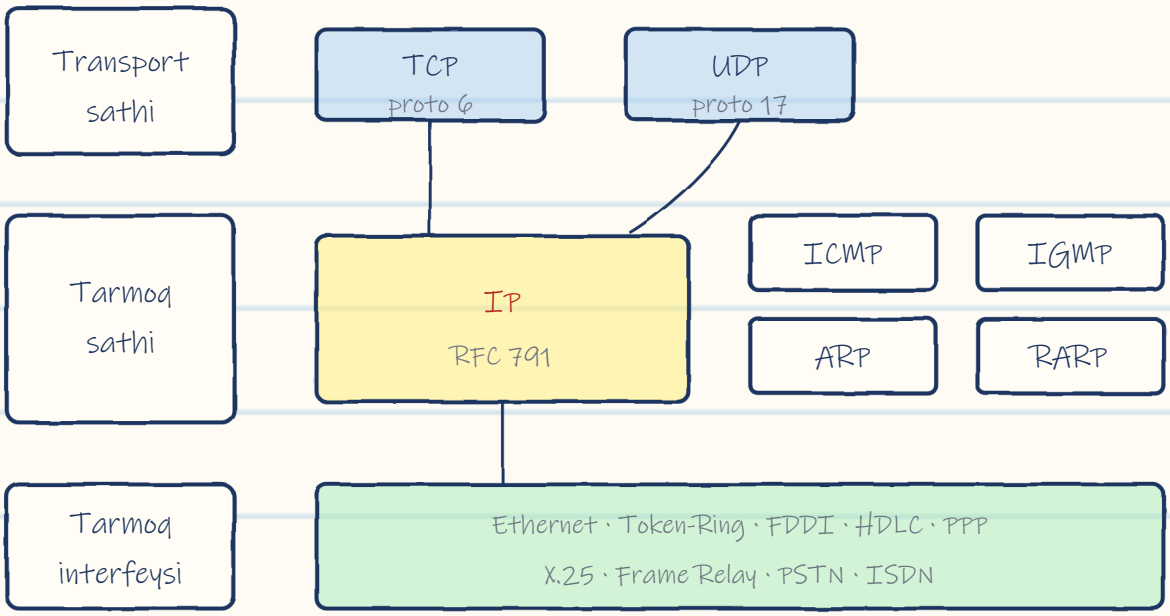
Asosiy atamalar

- IP manzil — qurilmaning tarmoqdagi manzili, 32 bit, 4 ta bayt
- Tarmoq va host qismi — manzilning boshi — tarmoq, oxiri — qurilma raqami
- Sinf (A/B/C) — manzil diapazoni: A — yirik, C — kichik tarmoqlar uchun
- Broadcast — host qismi butunlay 1 — tarmoqdagi hammaga xabar
- ARP — IP manzil bo'yicha MAC manzilni topish protokoli
- ICMP — xato va diagnostika xabarlari (ping shu protokolda)

## 1. Tarmoq sathi va IP

IP — TCP/IP tarmoq sathining asosiy protokoli, RFC 791. Ishonchsiz, ulanishsiz, best effort.

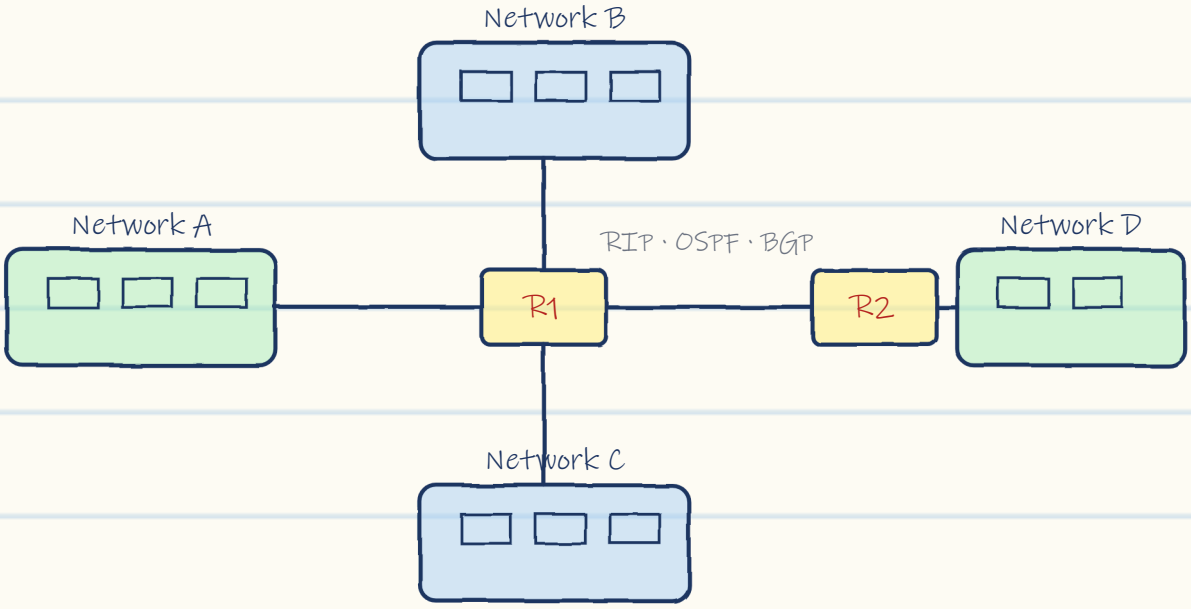
IP — tarmoq sathining yadrosi



- ARP — IP manzilni fizik (MAC) manzilga o'giradi
- RARP — fizik manzilni IP ga o'giradi
- ICMP — xato xabarlari, diagnostika · IGMP — multicast guruhlari

> Qoida: IP paket ichini o'qimaydi va holatini saqlamaydi. Ishonchlilik — TCP ning ishi.

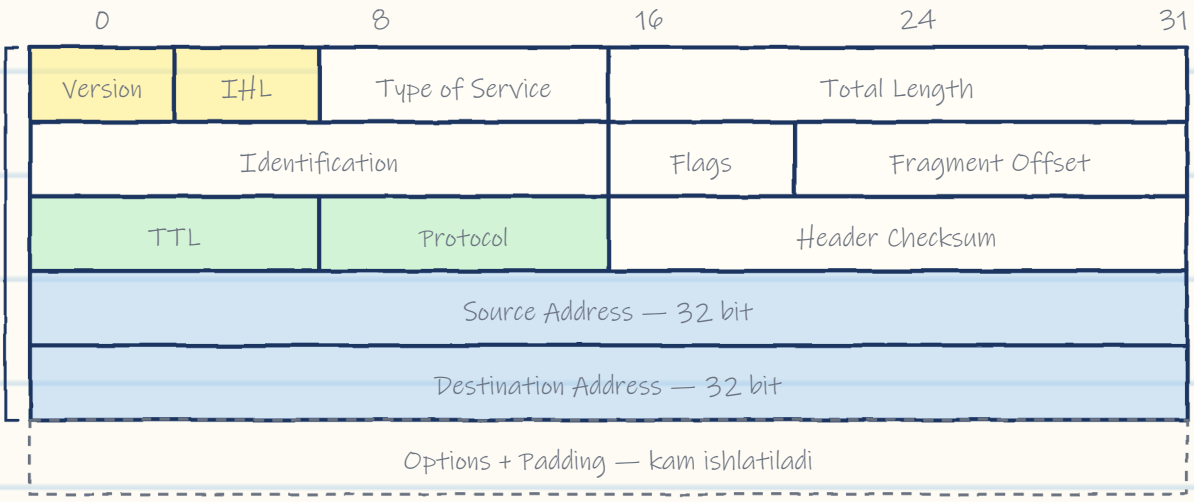
## 2. IP tarmoq tuzilishi




har segment = bitta link · router har interfeysi bilan boshqa tarmoqda

1-rasm. Paket hop-by-hop ketadi, har tugun o'z jadvalidan foydalanadi

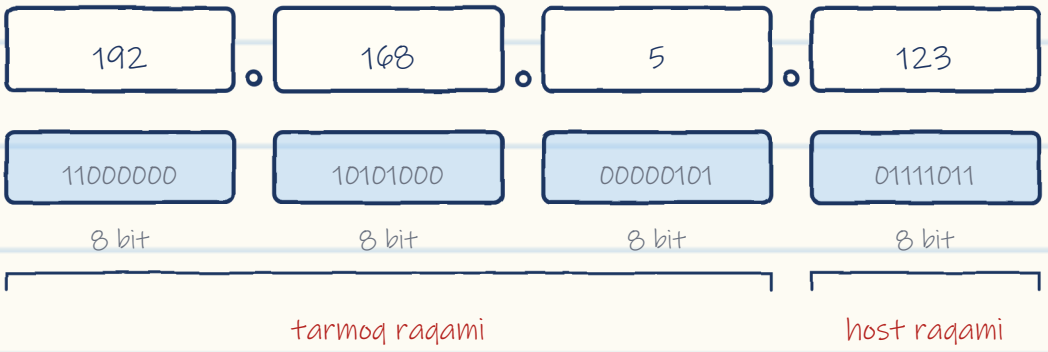
## 3. IP sarlavhasi




Options'siz sarlavha — 20 bayt

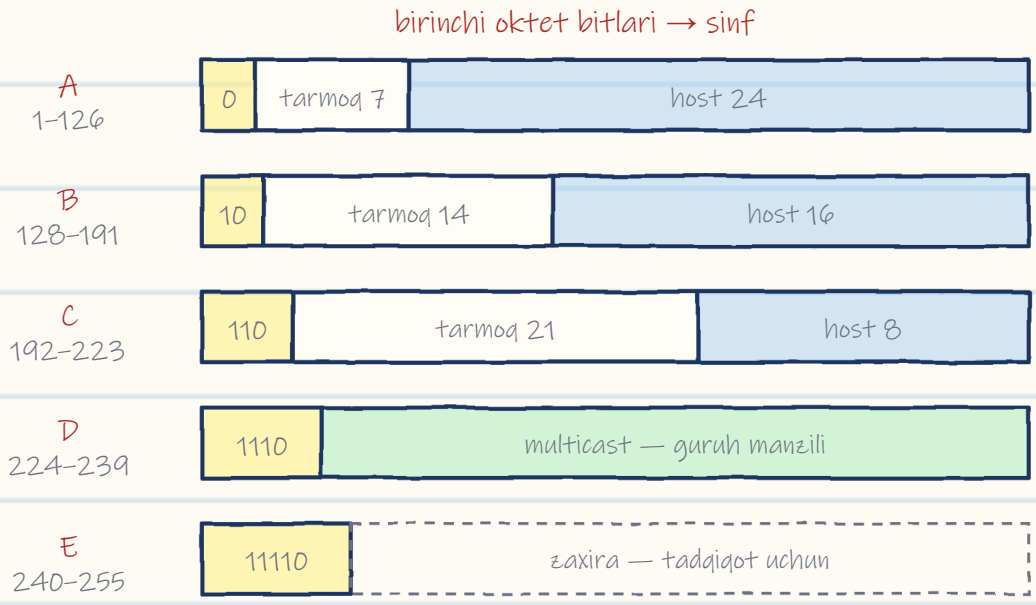
TTL har routerda −1 · 0 bo'lsa paket tashlanadi

## 4. IP manzil formati




32 bit · mantiqiy manzil, apparatdan mustaqil · $2^{32} \approx 4{,}3$ mlrd

## 5. Manzil sinflari

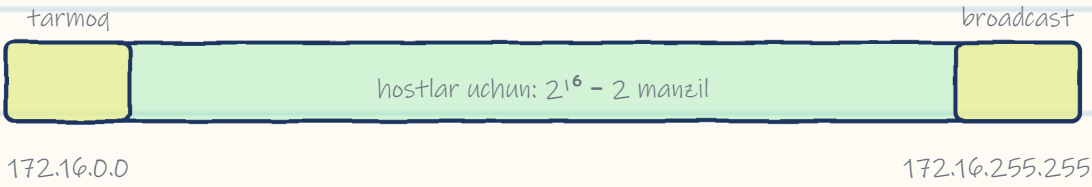



127 zaxira (loopback) · A tarmoqda $2^{24}$, B da $2^{16}$, C da $2^{8}$ manzil

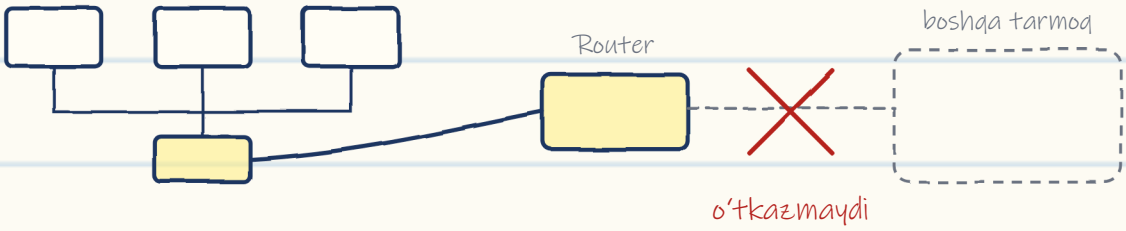
## 6. Maxsus manzillar

| Tarmoq | Host | Ma'nosi |
|---|---|---|
| har qanday | hammasi 0 | tarmoq manzili — 10.0.0.0 |
| har qanday | hammasi 1 | segment broadcast — 10.255.255.255 |
| hammasi 0 | hammasi 0 | 0.0.0.0 — hamma tarmoq, default marshrut |
| hammasi 1 | hammasi 1 | 255.255.255.255 — cheklangan broadcast |
| 127 | har qanday | loopback — 127.0.0.1 |

har segmentda 2 ta manzil band




broadcast — DIP 255.255.255.255

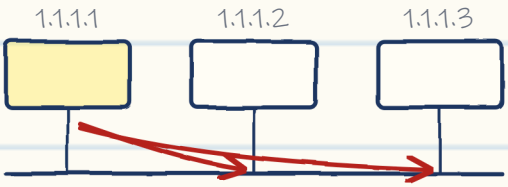



aks holda broadcast bo'roni — tarmoq qulaydi

Qoida: tarmoq va broadcast manzillari hostga berilmaydi.

## 7. ARP — IP dan MAC ga

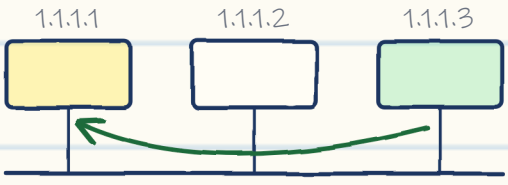
1) ARP so'rov — broadcast




IP = 1.1.1.3, MAC = ?

hammaga boradi

2) ARP javob — unicast




1.1.1.3 → 00E0.FC03.3333

faqat so'ragan hostga

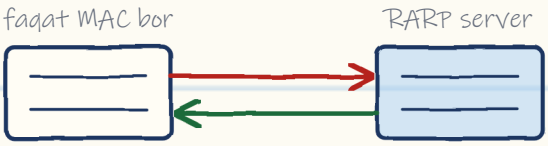
ARP jadval: dinamik — eskiradi · statik — qo'lda, ustun

Gratuitous ARP: o'z IP sini so'raydi — dublikat va MAC yangilanishi

Proxy ARP: router boshqa tarmoq o'rniga o'z MAC ini beradi

# 8. RARP va ICMP

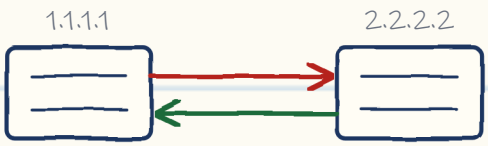
### RARP — disksiz ish stansiyasi




broadcast: IP = ?

unicast: IP = 1.1.1.1

### ICMP — ping




Echo Request — 8

Echo Reply — 0

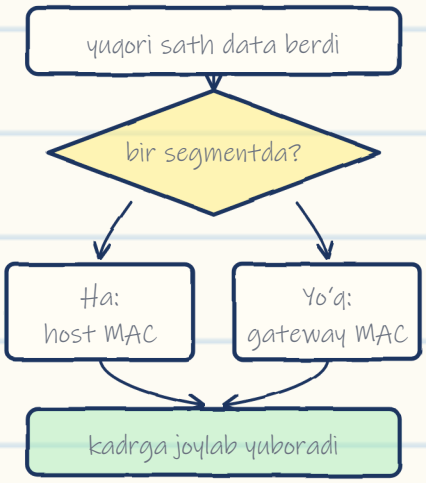
xato xabarlari: 3 — yetib bo'lmaydi · 11 — vaqt tugadi · 5 — redirect

4 — source quench · 12 — parametr xatosi · 13/14 — timestamp

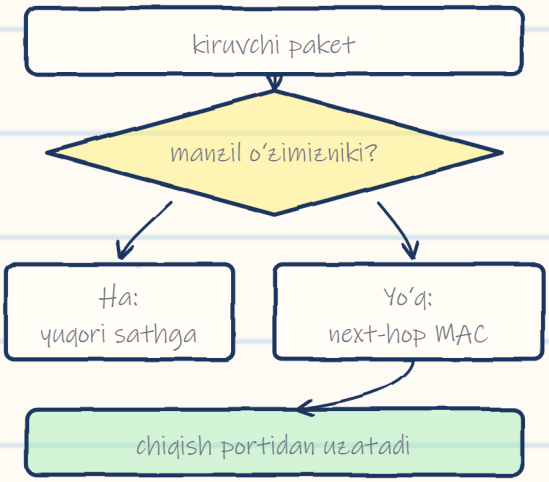
so'rov — broadcast, javob — unicast; resurs tejaladi

# 9. Paketni yetkazish

### Host yuboradi




### Router uzatadi




host qabul qiladi: o'z IP si · broadcast · a'zo bo'lgan multicast

## Xulosa

- 32 bitli IP = tarmoq raqami + host raqami
- Sinflar: A 1–126 · B 128–191 · C 192–223 · D multicast · E zaxira
- Sarlavha 20 bayt; TTL har routerda −1
- Boshqa tarmoqqa — default gateway orqali
- ARP: IP → MAC · RARP: MAC → IP · ICMP: xato va diagnostika

### Tekshirish

1. TCP va UDP protokol raqami? — 6 va 17
2. Sarlavha uzunligi? — 20 bayt (Options'siz)
3. 127.0.0.1 nima? — loopback
4. Nega −2? — tarmoq va broadcast manzillari
5. ARP so'rov qanday ketadi? — broadcast, javob unicast